ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success



PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Gravitation

English Medium

EXERCISE (O-1)

A. Gravitational Force

1. Two identical particles of combined mass M, placed in space with certain separation, are released. Interaction between the particles is only of gravitational in nature and there is no external force present. Acceleration of one particle with respect to the other when separation between them is R, has a magnitude :

(A) $\frac{GM}{2R^2}$

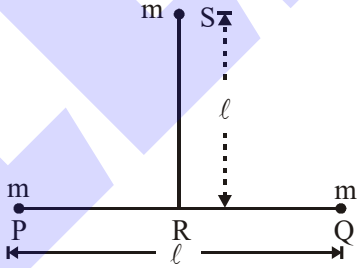
(B) $\frac{GM}{R^2}$

(C) $\frac{2GM}{R^2}$

(D) not possible to calculate due to lack of information

GR0133

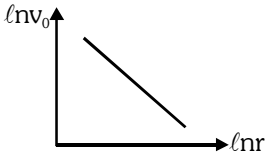
2. Two particles each of mass m are placed at points P and Q as shown in the figure. R is the mid-point of PQ = ℓ. The gravitational force on the third particle of mass m placed at point S on the perpendicular bisector of PQ at distance ℓ from R is:-



(A) $\frac{Gm^2}{\ell^2}$ (B) $\frac{16Gm^2}{5\ell^2}$ (C) $\frac{16Gm^2}{5\sqrt{5}\,\ell^2}$ (C) $\frac{4\sqrt{2}Gm^2}{5\,\ell^2}$

GR0134

3. If the law of gravitation be such that the force of attraction between two particles vary inversely as the $5/2^{th}$ power of their separation, then the graph of orbital velocity v_0 plotted against the distance r of a satellite from the earth's centre on a log-log scale is shown alongside. The slope of line will be-



(A) $-\frac{5}{4}$ (B) $-\frac{5}{2}$ (C) $-\frac{3}{4}$ (D) –1

GR0135

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4. The gravitational force of attraction between two bodies is F newtons. If the mass of each body and the distance between them are doubled, then the gravitational force between them in newton is
(A) 16 F (B) F/16 (C) F/4 (D) F

GR0136

5. Suppose the gravitational force varies inversely as the nth power of distance. Then, the time period of a planet in circular orbit of radius R around the sun will be proportional to

(A) R^n (B) $R^{\frac{n+1}{2}}$ (C) $R^{\frac{n-1}{2}}$ (D) R^{-n}

GR0137

6. If three uniform spheres, each having mass M and radius R, are kept in such a way that each touches the other two, the magnitude of the gravitational force on any sphere due to the other two is

(A) $\frac{GM^2}{4r^2}$ (B) $\frac{2GM^2}{r^2}$ (C) $\frac{2GM^2}{4r^2}$ (D) $\frac{\sqrt{3}GM^2}{4r^2}$

GR0138

B. Gravitational Intensity, Potential and Potential energy

7. An object is placed at a distance of R/2 from the centre of earth. Knowing mass is distributed uniformly, acceleration of that object due to gravity at that point is: (g = acceleration due to gravity on the surface of earth and R is the radius of earth)
(A) g (B) 2 g (C) g/2 (D) none of these

GR0139

8. A spherical shell has mass M and radius R. A point mass m/2 kept inside the shell at a distance R/2 from centre. Then force of attraction on the mass is:

(A) $\frac{2Gm^2}{R^2}$ (B) $\frac{Gm^2}{R^2}$ (C) $\frac{Gm^2}{2R}$ (D) zero

GR0140

9. At certain height h from surface of earth the value of g become 6.25% of its value at earth surface. The ratio of $\frac{h}{R}$ is. (R is radius of earth)
(A) 3 (B) 2 (C) 4 (D) 1

GR0141

10. How much deep inside the earth (radius R) should a man go, so that his weight becomes one-fourth of that on the earth's surface–

(A) $\frac{R}{4}$ (B) $\frac{R}{2}$ (C) $\frac{3R}{4}$ (D) None

GR0142

11. If the radius of the earth were to shrink by one percent, its mass remaining the same, the value of g on the earth's surface would
(A) increase by 0.5% (B) increase by 2%
(C) decrease by 0.5% (D) decrease by 2%

GR0143

12. The acceleration due to gravity (on earth) depends upon
(A) size of the body (B) gravitational mass of the body
(C) gravitational mass of the earth (D) none of the above factors

GR0144

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13. The average magnitude of intensity of the gravitational field on the surface of the earth in SI unit is about

(A) zero (B) 10 (C) 100 (D) more than 100

GR0145

14. If the earth stops rotating about its axis, the acceleration due to gravity will remain unchanged at

(A) equator (B) latitude 45° (C) latitude 60° (D) poles

GR0146

15. One goes from the centre of the earth to an altitude half the radius of the earth, where will the g be greatest ?

(A) centre of the earth

(B) At a depth half the radius of the earth

(C) At the surface of the earth

(D) At an altitude equal to half the radius of the earth.

GR0147

16. An iron sphere and an aluminium sphere, both of same radius are dropped from the top of a tower 100m high. At a height 40 m above the ground, both of them will have same.

(A) momentum (B) kinetic energy

(C) potential energy (D) acceleration

GR0148

17. The acceleration due to gravity increases by 0.5 % when we go from the equator to the poles. What will be the time period of the pendulum at the equator which beats seconds at the poles?

(A) 1.950 s (B) 1.995 s (C) 2.050 s (D) 2.005 s

GR0149

18. A thin rod of length L is bent to form a circle. Its mass is M. What force will act on the mass m placed at the centre of the circle ?

(A) $\frac{4\pi^2 GMm}{L^2}$ (B) $\frac{GMm}{4\pi^2 L^2}$ (C) $\frac{2\pi GMm}{L^2}$ (D) zero

GR0150

19. Two equal masses m and m are hung from a balance whose scale pans differ in vertical height by 'h'. The error in weighing in terms of density of the earth ρ is :

(A) $\pi G\rho mh$ (B) $\frac{1}{3}\pi G\rho mh$ (C) $\frac{8}{3}\pi G\rho mh$ (D) $\frac{4}{3}\pi G\rho mh$

GR0151

20. Two bodies of masses m and M are placed at distance d apart. What is the gravitational potential (V) at the position where the gravitational field due to them is zero is V :

(A) $V = -\frac{G}{d}(m+M)$ (B) $V = -\frac{G}{d}m$ (C) $V = -\frac{GM}{d}$ (D) $V = -\frac{G}{d}\left(\sqrt{m}+\sqrt{M}\right)^2$

GR0152

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21. Gravitational potential difference between a point on surface of planet and another point 10m above is 4J/kg. Considering gravitational field to be uniform, how much work is done in moving a mass of 2.0 kg from the surface to a point 5.0m above the surface?

(A) 0.40 J (B) 2.5 J (C) 4.0 J (D) 8.0 J

GR0153

22. Two concentric shells have mass M and m and their radii are R and r respectively, where $R > r$. What is the gravitational potential at their common centre ?

(A) $-\frac{GM}{R}$ (B) $-\frac{GM}{r}$ (C) $-G\left[\frac{M}{R}-\frac{m}{r}\right]$ (D) $-G\left[\frac{M}{R}+\frac{m}{r}\right]$

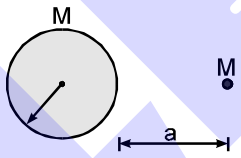
GR0154

23. The gravitational force between two particles with masses m and M, initially at rest at great separation, pulls them together. When their separation becomes d, then speed of either particle relative to the other will be :

(A) $\sqrt{G(M+m)/2d}$ (B) $\sqrt{G(M+m)/d}$ (C) $\sqrt{4G(M+m)/d}$ (D) $\sqrt{2G(M+m)/d}$

GR0155

24. A particle of mass M is at a distance a from surface of a thin spherical shell of equal mass and having radius a.



(A) Gravitational field and potential both are zero at centre of the shell.
(B) Gravitational field is zero not only inside the shell but at a point outside the shell also.
(C) Inside the shell, gravitational field alone is zero.
(D) Neither gravitational field nor gravitational potential is zero inside the shell.

GR0156

25. If escape velocity from earth is 11.2 km/s, Then escape velocity from a planet of mass as that of earth but of its one fourth radius

(A) 11.2 km/s (B) 22.4 km/s (C) 5.6 km/s (D) 44.8 km/s

GR0157

26. A tunnel is dug along the diameter of the earth (radius R and mass M). There is a particle of mass 'm' at the centre of the tunnel. The minimum velocity given to the particle so that it just reaches to the surface of the earth is:

(A) $\sqrt{\frac{GM}{R}}$ (B) $\sqrt{\frac{GM}{2R}}$

(C) $\sqrt{\frac{2GM}{R}}$ (D) it will reach with the help of negligible velocity

GR0158

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27. What is the increase in gravitational potential energy of an object of mass m raised from the surface of earth to a height equal to n times of earth radius ?

(A) $\left(\frac{n+1}{n}\right)mgR$ (B) $\left(\frac{n-1}{n}\right)mgR$ (C) $\left(\frac{n}{n-1}\right)mgR$ (D) $\left(\frac{n}{n+1}\right)mgR$

GR0159

28. The escape velocity for a planet is v_e. A particle starts from rest at a large distance from the planet, reaches the planet only under gravitational attraction, and passes through a smooth tunnel through its centre. Its speed at the centre of the planet will be-

(A) $\sqrt{1.5}v_e$ (B) $\frac{v_e}{\sqrt{2}}$ (C) v_e (D) zero

GR0160

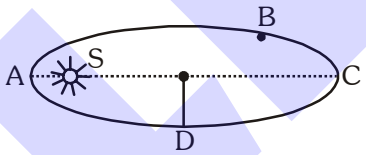
29. The mass of a spaceship is 1000 kg. It is to be launched from the earth's surface out into free space. The value of 'g' and 'R' (radius of earth) are 10 m/s^2 and 6400 km respectively. The required energy for this work will be :- **[AIEEE-2012]**

(A) 6.4×10^{10} Joules (B) 6.4×10^{11} Joules
(C) 6.4×10^{8} Joules (D) 6.4×10^{9} Joules

GR0053

C. Kepler's Law and Satellite motion

30. A planet revolves in elliptical orbit around the sun. The linear speed of the planet will be maximum at



(A) A (B) B (C) C (D) D

GR0161

31. A satellite is launched in the equatorial plane in such a way that it can transmit signals upto 60^0 latitude on the earth. Then the angular velocity of the satellite is :

(A) $\sqrt{\frac{GM}{8R^3}}$ (B) $\sqrt{\frac{GM}{2R^3}}$ (C) $\sqrt{\frac{GM}{4R^3}}$ (D) $\sqrt{\frac{3\sqrt{3}GM}{8R^3}}$

GR0162

32. A satellite is seen after each 8 hours over equator at a place on the earth when its sense of rotation is opposite to the earth. The time interval after which it can be seen at the same place when the sense of rotation of earth & satellite is same will be :

(A) 8 hours (B) 12 hours (C) 24 hours (D) 6 hours

GR0163

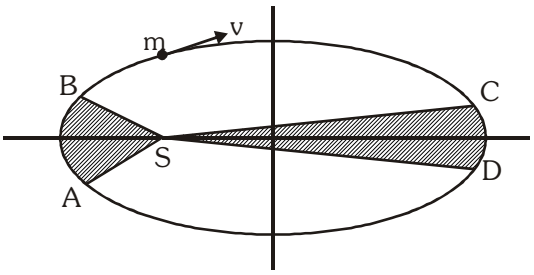
33. A satellite of mass m is in a circular orbit of radius 2R about the earth. How much energy is required to transfer it to a circular orbit of radius 4R :- (R = Radius of earth)

(A) $\frac{mgR}{8}$ (B) $\frac{mgR}{4}$ (C) $\frac{mgR}{2}$ (D) None of these

GR0164

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34. The figure shows elliptical orbit of a planet m about the sun S. The shaded area SCD is twice the shaded area SAB. If t_1 be the time for the planet to move from C to D and t_2 is the time to move from A to B, then :



(A) $t_1 = t_2$ (B) $t_1 = 8t_2$ (C) $t_1 = 4t_2$ (D) $t_1 = 2t_2$

GR0165

35. If the earth be at one half its present distance from the sun, number of days in the year will be nearly

(A) 129 (B) 30 (C) 200 (D) 60

GR0166

36. The period of a satellite in a circular orbit of radius R is T. What is the period of another satellite in a circular orbit of radius 4R ?

(A) 4T (B) T/8 (C) T/4 (D) 8 T

GR0167

37. Two satellites S and S' revolve around the earth at distances 3R and 6R from the centre of earth. Their periods of revolution will be in the ratio

(A) 1 : 2 (B) 2 : 1 (C) $1 : 2^{1.5}$ (D) $1 : 2^{0.67}$

GR0168

38. A satellite revolves around a planet in an elliptical orbit of minor and major axes a and b respectively. If T be the time period of the satellite, then T^2 is proportional to

(A) $\left(\frac{a+b}{2}\right)^3$ (B) $\left(\frac{a-b}{2}\right)^3$ (C) a^3 (D) b^3

GR0169

39. A geostationary satellite has an orbital period of

(A) 2 hours (B) 6 hours (C) 12 hours (D) 24 hours

GR0170

40. Imagine a light planet revolving around a very massive star in a circular orbit of radius r with a period of revolution T. If the gravitational force of attraction between the planet and the star is proportional to $r^{5/2}$, then the square of the time period will be proportional to

(A) r^3 (B) r^2 (C) $r^{2.5}$ (D) $r^{3.5}$

GR0171

41. Satellites A and B are orbiting around the earth in orbits of ratio R and 4R respectively. The ratio of their areal velocities is :

(A) 1 : 2 (B) 1 : 4 (C) 1 : 8 (D) 1 : 16

GR0033

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EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

A. Gravitational Force

1. If suddenly the gravitational force of attraction between earth and a satellite revolving around it becomes zero, then the satellite will- **[AIEEE-2002]**

(A) continue to move in its orbit with same velocity

(B) move tangentially to the original orbit with same velocity

(C) become stationary in its orbit

(D) move towards the earth

GR0035

B. Gravitational Intensity, Potential and Potential energy

2. If the distance between the centres of Earth and Moon is D and mass of Earth is 81 times that of Moon. At what distance from the centre of Earth gravitational field will be zero?

(A) $\frac{D}{2}$ (B) $\frac{2D}{3}$ (C) $\frac{4D}{5}$ (D) $\frac{9D}{10}$

GR0018

3. A hollow spherical shell is compressed to half its radius. The gravitational potential at the centre

(A) increases

(B) decreases

(C) remains same

(D) during the compression increases then returns at the previous value.

GR0020

4. Let ω be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles in absence of rotation of earth. An object weighed at the equator gives the same reading as a reading taken at a depth d below earth's surface at a pole (d<<R) The value of d is

(A) $\frac{\omega^2 R^2}{g}$ (B) $\frac{\omega^2 R^2}{2g}$ (C) $\frac{2\omega^2 R^2}{g}$ (D) $\frac{\sqrt{Rg}}{g}$

GR0021

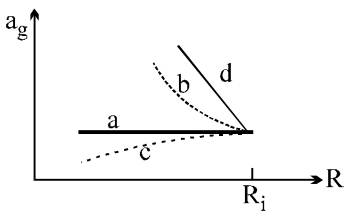
5. A spherical uniform planet is rotating about its axis. The velocity of a point on its equator is V. Due to the rotation of planet about its axis the acceleration due to gravity g at equator is 1/2 of g at poles. The escape velocity of a particle on the pole of planet in terms of V is

(A) $V_e = 2V$ (B) $V_e = V$ (C) $V_e = V/2$ (D) $V_e = \sqrt{3}\,V$

GR0025

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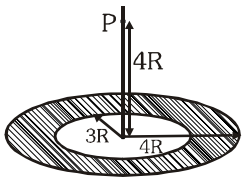
6. A (nonrotating) star collapses onto itself from an initial radius R_i with its mass remaining unchanged. Which curve in figure best gives the gravitational acceleration a_g on the surface of the star as a function of the radius of the star during the collapse?



(A) a (B) b
(C) c (D) d

GR0027

7. A thin uniform annular disc (see figure) of mass M has outer radius 4R and inner radius 3R. The work required to take a unit mass from point P on its axis to infinity is **[IIT-JEE 2010]**



(A) $\frac{2GM}{7R}\left(4\sqrt{2}-5\right)$ (B) $-\frac{2GM}{7R}\left(4\sqrt{2}-5\right)$ (C) $\frac{GM}{4R}$ (D) $\frac{2GM}{5R}\left(\sqrt{2}-1\right)$

GR0061

8. Time period of simple pendulum of infinite length oscillating near the surface of the earth is:
[M_e = mass of the earth, R = radius]

(A) $2\pi\sqrt{\frac{R^2}{GM_e}}$ (B) $2\pi\sqrt{\frac{R^3}{GM_e}}$ (C) $2\pi\sqrt{\frac{R}{GM_e}}$ (D) $2\pi\sqrt{\frac{R^4}{GM_e}}$

GR0172

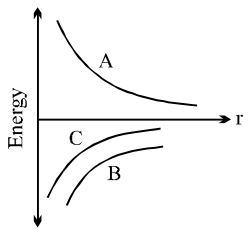
C. Kepler's Law and Satellite motion

9. The mass and diameter of a planet are twice those of earth. What will be the period of oscillation of a pendulum on this planet if it is a seconds pendulum on earth?

(A) $\sqrt{2}$ second (B) $2\sqrt{2}$ seconds (C) $\frac{1}{\sqrt{2}}$ second (D) $\frac{1}{2\sqrt{2}}$ second

GR0023

10. The figure shows the variation of energy with the orbit radius of a body in circular planetary motion. Find the correct statement about the curves A, B and C



(A) A shows the kinetic energy, B the total energy and C the potential energy of the system.
(B) C shows the total energy, B the kinetic energy and A the potential energy of the system.
(C) C and A are kinetic and potential energies respectively and B is the total energy of the system.
(D) A and B are kinetic and potential energies and C is the total energy of the system.

GR0029

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11. A satellite of mass 5M orbits the earth in a circular orbit. At one point in its orbit, the satellite explodes into two pieces, one of mass M and the other of mass 4M. After the explosion the mass M ends up travelling in the same circular orbit, but in opposite direction. After explosion the mass 4M is :-

(A) In a circular orbit

(B) unbound

(C) elliptical orbit

(D) data is insufficient to determine the nature of the orbit.

GR0030

12. A satellite can be in a geostationary orbit around earth at a distance r from the centre. If the angular velocity of earth about its axis doubles, a satellite can now be in a geostationary orbit around earth if its distance from the centre is :-

(A) $\frac{r}{2}$ (B) $\frac{r}{2\sqrt{2}}$ (C) $\frac{r}{(4)^{1/3}}$ (D) $\frac{r}{(2)^{1/3}}$

GR0031

13. An earth satellite is moved from one stable circular orbit to another larger and stable circular orbit. The following quantities increase for the satellite as a result of this change:-

(A) gravitational potential energy (B) angular velocity

(C) linear orbital velocity (D) centripetal acceleration

GR0032

MULTIPLE CORRECT TYPE QUESTIONS

C. Kepler's Law and Satellite motion

14. A communications Earth satellite

(A) goes round the earth from east to west

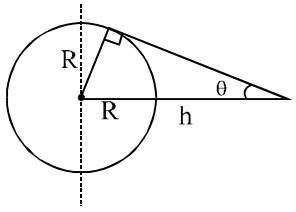
(B) can be in the equatorial plane only

(C) can be vertically above any place on the earth

(D) goes round the earth from west to east

GR0037

15. A geostationary satellite is at a height h above the surface of earth. If earth radius is R



(A) The minimum colatitude on earth upto which the satellite can be used for communication is $\sin^{-1}\left(R/R+h\right)$.

(B) The maximum latitudes on earth upto which the satellite can be used for communication is $\cos^{-1}\left(R/R+h\right)$.

(C) The area on earth escaped from this satellite is given as $2\pi R^2 (1 + \sin\theta)$

(D) The area on earth escaped from this satellite is given as $2\pi R^2 (1 + \cos\theta)$

GR0045

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16. A double star is a system of two stars of masses m and 2m, rotating about their centre of mass only under their mutual gravitational attraction. If r is the separation between these two stars then their time period of rotation about their centre of mass will be proportional to :

(A) $r^{3/2}$ (B) r (C) $m^{1/2}$ (D) $m^{-1/2}$

GR0048

17. Two spherical planets P and Q have the same uniform density ρ, masses M_P and M_Q, and surface areas A and 4A, respectively. A spherical planet R also has uniform density ρ and its mass is $(M_P + M_Q)$. The escape velocities from the planets P, Q and R, are V_P, V_Q and V_R, respectively. Then

[IIT-JEE 2012]

(A) $V_Q > V_R > V_P$ (B) $V_R > V_Q > V_P$ (C) $V_R/V_P = 3$ (D) $V_P/V_Q = 1/2$

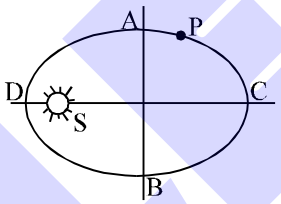
GR0063

COMPREHENSION TYPE QUESTIONS

C. Kepler's Law and Satellite motion

Paragraph for Question No. 18 and 19

Figure shows the orbit of a planet P around the sun S. AB and CD are the minor and major axes of the ellipse.



18. If t_1 is the time taken by the planet to travel along ACB and t_2 the time along BDA, then

(A) $t_1 = t_2$ (B) $t_1 > t_2$

(C) $t_1 < t_2$ (D) nothing can be concluded

GR0050

19. If U is the potential energy and K kinetic energy then $|U| > |K|$ at

(A) Only D (B) Only C (C) both D & C (D) neither D nor C

GR0050

MATRIX MATCH TYPE QUESTIONS

C. Kepler's Law and Satellite motion

20. In elliptical orbit of a planet, as the planet moves from apogee position to perigee position,

	Column–I		**Column–II**
(A)	Speed of planet	(P)	Remains same
(B)	Distance of planet from centre of Sun	(Q)	Decreases
(C)	Potential energy	(R)	Increases
(D)	Angular momentum about centre of Sun	(S)	Can not say

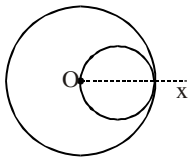
GR0040

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EXERCISE (S)

B. Gravitational Intensity, Potential and Potential energy

1. A sphere of radius R has its centre at the origin. It has a uniform mass density ρ_o except that there is a spherical hole of radius r = R/2 whose centre is at x = R/2 as in fig. (a) Find gravitational field at points on the axis for |x| > R (b) Show that the gravitational field inside the hole is uniform, find its magnitude and direction.



GR0008

2. A particle is fired vertically from the surface of the earth with a velocity $k\upsilon_e$, where υ_e is the escape velocity and k < 1. Neglecting air resistance and assuming earth's radius as R_e. Calculate the height to which it will rise from the surface of the earth.

GR0001

3. Calculate the distance from the surface of the earth at which above and below the surface acceleration due to gravity is the same.

GR0002

C. Kepler's Law and Satellite motion

4. A satellite close to the earth is in orbit above the equator with a period of rotation of 1.5 hours. If it is above a point P on the equator at some time, it will be above P again after time_____.

GR0004

5. A satellite is moving in a circular orbit around the earth. The total energy of the satellite is $E = -2 \times 10^5$J. The amount of energy to be imparted to the satellite to transfer it to a circular orbit where its potential energy is $U = -2 \times 10^5$J is equal to _____.

GR0005

6. A satellite of mass m is orbiting the earth in a circular orbit of radius r. It starts losing energy due to small air resistance at the rate of C J/ s. Then the time taken for the satellite to reach the earth is ____.

GR0006

7. A pair of stars rotates about a common center of mass. One of the stars has a mass M which is twice as large as the mass m of the other. Their centres are at a distance d apart, d being large compared to the size of either star. (a) Derive an expression for the period of rotation of the stars about their common centre of mass in terms of d,m, G. (b) Compare the angular momentum of the two stars about their common centre of mass by calculating the ratio L_m/L_M. (c) Compare the kinetic energies of the two stars by calculating the ratio K_m/K_M.

GR0007

8. A satellite of mass m moves in an elliptic orbit around a planet of mass M, so that its maximum and minimum distances from the planet are r_1 and r_2 respectively. Find the angular momentum L of this satellite relative to the centre of planet.

GR0173

9. A binary star consists of two stars A (mass 2.2 M_S) and B (mass 11 M_S), where M_S is the mass of the sun. They are separated by distance d and are rotating about their centre of mass, which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of mass is :- **[IIT-JEE 2010]**

GR0060

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EXERCISE (JM)

1. What is the minimum energy required to launch a satellite of mass m from the surface of a planet of mass M and radius R in a circular orbit at an altitude of 2R ? **[JEE-Main 2013]**

(1) $\frac{5GmM}{6R}$ (2) $\frac{2GmM}{3R}$ (3) $\frac{GmM}{2R}$ (4) $\frac{GmM}{3R}$

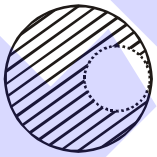
GR0054

2. Four particles, each of mass M and equidistant from each other, move along a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is : **[JEE-Main 2014]**

(1) $\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$ (2) $\frac{1}{2}\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$ (3) $\sqrt{\frac{GM}{R}}$ (4) $\sqrt{2\sqrt{2}\frac{GM}{R}}$

GR0055

3. From a solid sphere of mass M and radius R, a spherical portion of radius $\frac{R}{2}$ is removed, as shown in the figure. Taking gravitational potential V = 0 at r = ∞, the potential at the centre of the cavity thus formed is : (G = *gravitational constant*) **[JEE-Main 2015]**



(1) $\frac{-2GM}{3R}$ (2) $\frac{-2GM}{R}$ (3) $\frac{-GM}{2R}$ (4) $\frac{-GM}{R}$

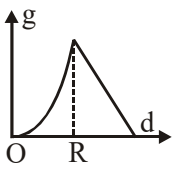
GR0056

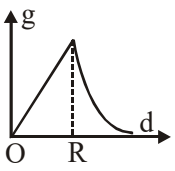
4. A satellite is reolving in a circular orbit at a height 'h' from the earth's surface (radius of earth R ; h << R). The minimum increase in its orbital velocity required, so that the satellite could escape from the earth's gravitational field, is close to : (Neglect the effect of atmosphere). **[JEE-Main 2016]**

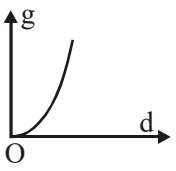
(1) $\sqrt{gR}\left(\sqrt{2}-1\right)$ (2) $\sqrt{2gR}$ (3) $\sqrt{gR}$ (4) $\sqrt{gR/2}$

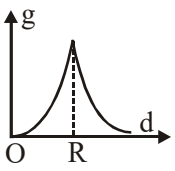
GR0057

5. The variation of acceleration due to gravity g with distance d from centre of the earth is best represented by (R = Earth's radius) :- **[JEE-Main 2017]**

(1) 

(2) 

(3) 

(4) 

GR0058

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SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

6. Two stars of masses 3×10^{31} kg each, and at distance 2×10^{11}m rotate in a plane about their common centre of mass O. A meteorite passes through O moving perpendicular to the star's rotation plane. In order to escape from the gravitational field of this double star, the minimum speed that meteorite should have at O is : (Take Gravitational constant $G = 6.67 \times 10^{-11}$ $Nm^2 kg^{-2}$)

[JEE-Main-2019_Jan]

(1) 1.4×10^5 m/s (2) 24×10^4 m/s (3) 3.8×10^4 m/s (4) 2.8×10^5 m/s

GR0096

7. A satellite is moving with a constant speed v in circular orbit around the earth. An object of mass 'm' is ejected from the satellite such that it just escapes from the gravitational pull of the earth. At the time of ejection, the kinetic energy of the object is : **[JEE-Main-2019_Jan]**

(1) $\frac{3}{2}mv^2$ (2) mv^2 (3) $2mv^2$ (4) $\frac{1}{2}mv^2$

GR0097

8. A straight rod of length L extends from x = a to x=L + a. The gravitational force is exerts on a point mass 'm' at x = 0, if the mass per unit length of the rod is $A + Bx^2$, is given by:

[JEE-Main-2019_Jan]

(1) $Gm\left[A\left(\frac{1}{a+L}-\frac{1}{a}\right)-BL\right]$ (2) $Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)+BL\right]$

(3) $Gm\left[A\left(\frac{1}{a+L}-\frac{1}{a}\right)+BL\right]$ (4) $Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)-BL\right]$

GR0098

9. A satellite of mass M is in a circular orbit of radius R about the centre of the earth. A meteorite of the same mass, falling towards the earth, collides with the satellite completely inelastically. The speeds of the satellite and the meteorite are the same, just before the collision. The subsequent motion of the combined body will be : **[JEE-Main-2019_Jan]**

(1) in a circular orbit of a different radius (2) in the same circular orbit of radius R

(3) in an elliptical orbit (4) such that it escapes to infinity

GR0099

10. A rocket has to be launched from earth in such a way that it never returns. If E is the minimum energy delivered by the rocket launcher, what should be the minimum energy that the launcher should have if the same rocket is to be launched from the surface of the moon ? Assume that the density of the earth and the moon are equal and that the earth's volume is 64 times the volume of the moon :-

[JEE-Main-2019_April]

(1) $\frac{E}{4}$ (2) $\frac{E}{16}$ (3) $\frac{E}{32}$ (4) $\frac{E}{64}$

GR0100

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11. A test particle is moving in a circular orbit in the gravitational field produced by a mass density $\rho(r) = \frac{K}{r^2}$. Identify the correct relation between the radius R of the particle's orbit and its period T :

[JEE-Main-2019_April]

(1) T/R^2 is a constant (2) TR is a constant

(3) T^2/R^3 is a constant (4) T/R is a constant

GR0101

12. A spaceship orbits around a planet at a height of 20 km from its surface. Assuming that only gravitational field of the planet acts on the spaceship, what will be the number of complete revolutions made by the spaceship in 24 hours around the planet ? [Given : Mass of planet = 8×10^{22} kg ; Radius of planet = 2×10^6 m, Gravitational constant G = 6.67×10^{-11} Nm^2/kg^2]

[JEE-Main-2019_April]

(1) 9 (2) 11 (3) 13 (4) 17

GR0102

13. A satellite of mass m is launched vertically upwards with an initial speed u from the surface of the earth. After it reaches height R (R = radius of the earth), it ejects a rocket of mass $\frac{m}{10}$ so that subsequently the satellite moves in a circular orbit. The kinetic energy of the rocket is (G is the gravitational constant; M is the mass of the earth): **[JEE-Main-2020_Jan]**

(1) $\frac{m}{20}\left(u - \sqrt{\frac{2GM}{3R}}\right)^2$ (2) $5m\left(u^2 - \frac{119}{200}\frac{GM}{R}\right)$

(3) $\frac{3m}{8}\left(u + \sqrt{\frac{5GM}{6R}}\right)^2$ (4) $\frac{m}{20}\left(u^2 + \frac{113}{200}\frac{GM}{R}\right)$

GR0103

14. A body A of mass m is moving in a circular orbit of radius R about a planet. Another body B of mass $\frac{m}{2}$ collides with A with a velocity which is half $\left(\frac{\vec{v}}{2}\right)$ the instantaneous velocity $\vec{v}$ of A. The collision is completely inelastic. Then, the combined body : **[JEE-Main-2020_Jan]**

(1) starts moving in an elliptical orbit around the planet.

(2) continues to move in a circular orbit

(3) Falls vertically downwards towards the planet

(4) Escapes from the Planet's Gravitational field.

GR0104

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15. The mass density of a spherical galaxy varies as $\frac{K}{r}$ over a large distance 'r' from its centre. In that region, a small star is in a circular orbit of radius R. Then the period of revolution, T depends on R as: **[JEE-Main-2020_Sep]**

(1) $T \propto R$ (2) $T^2 \propto \frac{1}{R^3}$ (3) $T^2 \propto R$ (4) $T^2 \propto R^3$

GR0105

16. A satellite is moving in a low nearly circular orbit around the earth. Its radius is roughly equal to that of the earth's radius R_e. By firing rockets attached to it, its speed is instantaneously increased in the direction of its motion so that is become $\sqrt{\frac{3}{2}}$ times larger. Due to this the farthest distance from the centre of the earth that the satellite reaches is R, value of R is : **[JEE-Main-2020_Sep]**

(1) $4R_e$ (2) $3R_e$ (3) $2R_e$ (4) $2.5R_e$

GR0106

17. On the x-axis and a dsitance x from the origin, the gravitational field due to a mass distribution is given by $\frac{Ax}{(x^2+a^2)^{3/2}}$ in the x-direction. The magnitu **[JEE-Main-2020_Sep]**

(1) $\frac{A}{(x^2+a^2)^{1/2}}$ (2) $\frac{A}{(x^2+a^2)^{3/2}}$ (3) $A(x^2+a^2)^{3/2}$ (4) $A(x^2+a^2)^{1/2}$

GR0107

18. A body weighs 49 N on a spring balance at the north pole. What will be its weight recorded on the same weighing machine, if it is shifted to the equator ?

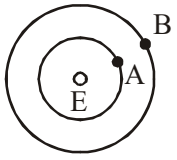
(Use $g = \frac{GM}{R^2} = 9.8\ ms^{-2}$ and radius of earth, R = 6400 km.] **[JEE-Main-2021_Feb]**

(1) 49 N (2) 48.83 N

(3) 49.83 N (4) 49.17 N

GR0174

19. Two satellites A and B of masses 200kg and 400kg are revolving round the earth at height of 600 km and 1600 km respectively. If T_A and T_B are the time periods of A and B respectively then the value of $T_B - T_A$: [Given : radius of earth = 6400km, mass of earth = 6×10^{24} kg] **[JEE-Main-2021_Feb]**



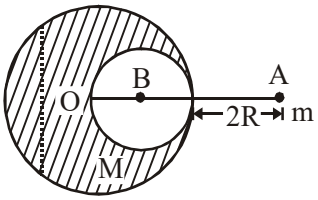
(1) 1.33×10^3 s (2) 3.33×10^2 s

(3) 4.24×10^3 s (4) 4.24×10^2 s

GR0175

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20. A solid sphere of radius R gravitationally attracts a particle placed at 3R form its centre with a force F_1. Now a spherical cavity of radius $\left(\frac{R}{2}\right)$ is made in the sphere (as shown in figure) and the force becomes F_2. The value of $F_1 : F_2$ is : **[JEE-Main-2021_Feb]**



(1) 25 : 36 (2) 36 : 25 (3) 50 : 41 (4) 41 : 50

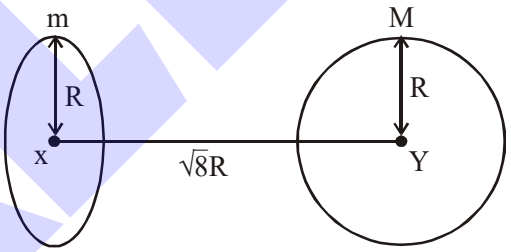
GR0176

21. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : The escape velocities of planet A and B are same. But A and B are of unequal mass.
Reason R : The product of their mass and radius must be same, $M_1R_1 = M_2R_2$
In the light of the above statements, choose the most appropriate answer from the options given below : **[JEE-Main-2021_Feb]**

(1) Both A and R are correct but R is NOT the correct explanation of A
(2) A is correct but R is not correct
(3) Both A and R are correct and R is the correct explanation of A
(4) A is not correct but R is correct

GR0177

22. Find the gravitational force of attraction between the ring and sphere as shown in the diagram, where the plane of the ring is perpendicular to the line joining the centres. If $\sqrt{8}\,R$ is the distance between the centres of a ring (of mass 'm') and a sphere (mass 'M') where both have equal radius 'R'.
[JEE-Main-2021_Feb]



(1) $\frac{\sqrt{8}}{9}\cdot\frac{GmM}{R}$ (2) $\frac{2\sqrt{2}}{3}\cdot\frac{GMm}{R^2}$ (3) $\frac{1}{3\sqrt{8}}\cdot\frac{GMm}{R^2}$ (4) $\frac{\sqrt{8}}{27}\cdot\frac{GmM}{R^2}$

GR0178

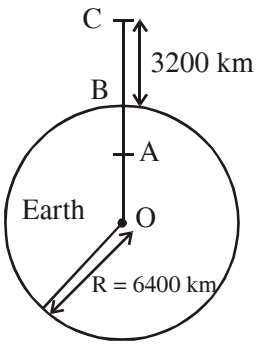
23. Assume that a tunnel is dug along a chord of the earth, at a perpendicular distance (R/2) from the earth's centre, where 'R' is the radius of the Earth. The wall of the tunnel is frictionless. If a particle is released in this tunnel, it will execute a simple harmonic motion with a time period : **[JEE-Main-2021_Feb]**

(1) $\frac{2\pi R}{g}$ (2) $\frac{g}{2\pi R}$ (3) $\frac{1}{2\pi}\sqrt{\frac{g}{R}}$ (4) $2\pi\sqrt{\frac{R}{g}}$

GR0179

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24. In the reported figure of earth, the value of acceleration due to gravity is same at point A and C but it is smaller than that of its value at point B (surface of the earth). The value of OA : AB will be x : y. The value of x is ..... **[JEE-Main-2021_Feb]**



GR0180

25. If one wants to remove all the mass of the earth to infinity in order to break it up completely. The amount of energy that needs to be supplied will be $\frac{x}{5}\frac{GM^2}{R}$ where x is ____ (Round off to the Nearest Integer) (M is the mass of earth, R is the radius of earth, G is the gravitational constant)

[JEE-Main-2021_March]

GR0181

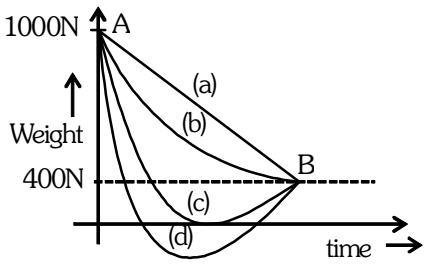
26. If the angular velocity of earth's spin is increased such that the bodies at the equator start floating, the duration of the day would be approximately :
(Take : g = 10 ms^{-2}, the radius of earth, R = 6400 × 10^3 m, Take π = 3.14)

[JEE-Main-2021_March]

(1) 60 minutes
(2) does not change
(3) 1200 minutes
(4) 84 minutes

GR0182

27. A person whose mass is 100 kg travels from Earth to Mars in a spaceship. Neglect all other objects in sky and take acceleration due to gravity on the surface of the Earth and Mars as 10 m/s^2 and 4 m/s^2 respectively. Identify from the below figures, the curve that fits best for the weight of the passenger as a function of time. **[JEE-Main-2021_July]**



(1) (c) (2) (a) (3) (d) (4) (b)

GR0183

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28. A body is projected vertically upwards from the surface of earth with a velocity sufficient enough to carry it to infinity. The time taken by it to reach height h is _____ S. **[JEE-Main-2021_July]**

(1) $\sqrt{\frac{R_e}{2g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$

(2) $\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$

(3) $\frac{1}{3}\sqrt{\frac{R_e}{2g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$

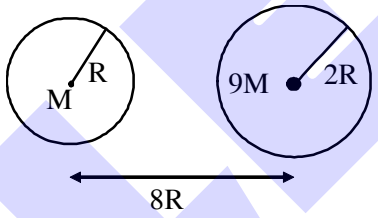
(4) $\frac{1}{3}\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$

GR0184

29. Suppose two planets (spherical in shape) of radii R and 2R, but mass M and 9 M respectively have a centre to centre separation 8 R as shown in the figure. A satellite of mass 'm' is projected from the surface of the planet of mass 'M' directly towards the centre of the second planet. The minimum speed 'v' required for the satellite to reach the surface of the second planet is $\sqrt{\frac{a}{7}\frac{GM}{R}}$ then the value of 'a' is _____.

[Given : The two planets are fixed in their position] **[JEE-Main-2021_July]**

GR0185



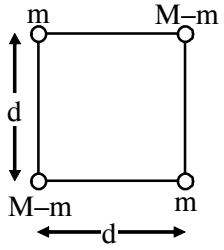
30. The planet Mars has two moons, if one of them has a period 7 hours, 30 minutes and an orbital radius of 9.0×10^3 km. Find the mass of Mars.

$\left\{\text{Given } \frac{4\pi^2}{G}=6\times10^{11}\ N^{-1}m^{-2}kg^2\right\}$ **[JEE-Main-2021_July]**

(1) 5.96×10^{19} kg (2) 3.25×10^{21} kg (3) 7.02×10^{25} kg (4) 6.00×10^{23} kg

GR0186

31. A body of mass (2M) splits into four masses {m, M – m, m, M – m}, which are rearranged to form a square as shown in the figure. The ratio of $\frac{M}{m}$ for which, the gravitational potential energy of the system becomes maximum is x : 1. The value of x is ..... . **[JEE-Main-2021_Aug]**



GR0187

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32. The masses and radii of the earth and moon are (M_1, R_1) and (M_2, R_2) respectively. Their centres are at a distance 'r' apart. Find the minimum escape velocity for a particle of mass 'm' to be projected from the middle of these two masses: **[JEE-Main-2021_Aug]**

(1) $V = \frac{1}{2}\sqrt{\frac{4G(M_1+M_2)}{r}}$ (2) $V = \sqrt{\frac{4G(M_1+M_2)}{r}}$

(3) $V = \frac{1}{2}\sqrt{\frac{2G(M_1+M_2)}{r}}$ (4) $V = \frac{\sqrt{2G}(M_1+M_2)}{r}$

GR0188

33. The approximate height from the surface of earth at which the weight of the body becomes $\frac{1}{3}$ of its weight on the surface of earth is : [Radius of earth R = 6400 km and $\sqrt{3} = 1.732$] **[JEE-Main-2022_June]**

(A) 3840 km (B) 4685 km

(C) 2133 km (D) 4267 km

GR0189

34. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : If we move from poles to equator, the direction of acceleration due to gravity of earth always points towards the center of earth without any variation in its magnitude.
Reason R : At equator, the direction of acceleration due to the gravity is towards the center of earth.
In the light of above statements, choose the correct answer from the options given below : **[JEE-Main-2022_June]**

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is NOT the correct explanation of A.
(C) A is true but R is false
(D) A is false but R is true

GR0190

35. Given below are two statements : **[JEE-Main-2022_June]**
Statement I : The law of gravitation holds good for any pair of bodies in the universe.
Statement II : The weight of any person becomes zero when the person is at the centre of the earth.
In the light of the above statements, choose the correct answer from the options given below.
(A) Both statement I and Statement II are true
(B) Both statement I and Statement II are false
(C) Statement I is true but Statement II are false
(D) Statement I is false but Statement II is true

GR0191

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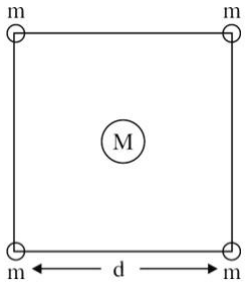
36. The distance of the Sun from earth is 1.5×10^{11} m and its angular diameter is (2000) s when observed from the earth. The diameter of the Sun will be : **[JEE-Main-2022_June]**

(A) 2.45×10^{10} m (B) 1.45×10^{10} m

(C) 1.45×10^{9} m (D) 0.14×10^{9} m

GR0192

37. Four spheres each of mass m form a square of side d (as shown in figure). A fifth sphere of mass M is situated at the centre of square. The total gravitational potential energy of the system is : **[JEE-Main-2022_June]**



(A) $-\frac{Gm}{d}\left[(4+\sqrt{2})m+4\sqrt{2}M\right]$ (B) $-\frac{Gm}{d}\left[(4+\sqrt{2})M+4\sqrt{2}m\right]$

(C) $-\frac{Gm}{d}\left[3m^2+4\sqrt{2}M\right]$ (D) $-\frac{Gm}{d}\left[6m^2+4\sqrt{2}M\right]$

GR0193

38. The escape velocity of a body on a planet 'A' is 12 kms^{-1}. The escape velocity of the body on another planet 'B', whose density is four times and radius is half of the planet 'A', is : **[JEE-Main-2022_June]**

(A) 12 kms^{-1} (B) 24 kms^{-1} (C) 36 kms^{-1} (D) 6 kms^{-1}

GR0194

39. The percentage decrease in the weight of a rocket, when taken to a height of 32 km above the surface of earth will, be : (Radius of earth = 6400km) **[JEE-Main-2022_July]**

(A) 1 % (B) 3% (C) 4% (D) 0.5%

GR0195

40. A body of mass m is projected with velocity λv_e in vertically upward direction from the surface of the earth into space. It is given that v_e is escape velocity and $\lambda < 1$. If air resistance is considered to the negligible, then the maximum height from the centre of earth, to which the body can go, will be (R : radius of earth) **[JEE-Main-2022_July]**

(A) $\frac{R}{1+\lambda^2}$ (B) $\frac{R}{1-\lambda^2}$ (C) $\frac{R}{1-\lambda}$ (D) $\frac{\lambda^2 R}{1-\lambda^2}$

GR0196

41. Assume there are two identical simple pendulum. Clock-1 is placed on the earth and Clock-2 is placed on a space station located at a height h above the earth surface. Clock-1 and Clock-2 operate at time periods 4s and 6s respectively. Then the value of h is - **[JEE-Main-2022_July]**
(consider radius of earth R_E = 6400 km and g on earth 10 m/s^2)

(A) 1200 km (B) 1600 km (C) 3200 km (D) 4800 km

GR0197

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EXERCISE (JA)

1. A planet of radius $R = \frac{1}{10} \times$ (radius of Earth) has the same mass density as Earth. Scientists dig a well of depth $\frac{R}{5}$ on it and lower a wire of the same length and of linear mass density 10^{-3} kgm^{-1} into it. If the wire is not touching anywhere, the force applied at the top of the wire by a person holding it in place is (take the radius of Earth = 6×10^6 m and the acceleration due to gravity on Earth is 10 ms^{-2}) **[JEE-Advance 2014]**

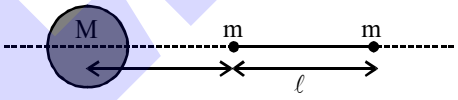
(A) 96 N (B) 108 N (C) 120 N (D) 150 N

GR0064

2. A bullet is fired vertically upwards with velocity v from the surface of a spherical planet. When it reaches its maximum height, its acceleration due to the planet's gravity is 1/4th of its value at the surface of the planet. If the escape velocity from the planet is $v_{esc} = v\sqrt{N}$, then the value of N is (ignore energy loss due to atmosphere) **[JEE-Advance 2015]**

GR0065

3. A large spherical mass M is fixed at one position and two identical point masses m are kept on a line passing through the centre of M (see figure). The point masses are connected by a rigid massless rod of length ℓ and this assembly is free to move along the line connecting them. All three masses interact only through their mutual gravitational interaction. When the point mass nearer to M is at a distance $r = 3\ell$ from M, the tension in the rod is zero for $m = k\left(\frac{M}{288}\right)$. The value of k is : **[JEE-Advance 2015]**



GR0066

4. A spherical body of radius R consists of a fluid of constant density and is in equilibrium under its own gravity. If P(r) is the pressure at r(r < R), then the correct option(s) is(are) :- **[JEE-Advance 2015]**

(A) $P(r = 0) = 0$ (B) $\frac{P(r = 3R/4)}{P(r = 2R/3)} = \frac{63}{80}$ (C) $\frac{P(r = 3R/5)}{P(r = 2R/5)} = \frac{16}{21}$ (D) $\frac{P(r = R/2)}{P(r = R/3)} = \frac{20}{27}$

GR0067

5. A rocket is launched normal to the surface of the Earth, away from the Sun, along the line joining the sun and the Earth. The Sun is 3×10^5 times heavier than the Earth and is at a distance 2.5×10^4 times larger than the radius of the Earth. The escape velocity from Earth's gravitational field is $v_e = 11.2$ km s^{-1}. The minimum initial velocity (v_s) required for the rocket to be able to leave the *Sun-Earth system* is closest to (*Ignore the rotation and revolution of the Earth and the presence of any other planet)* **[JEE-Advance 2017]**

(A) $v_s = 22$ km s^{-1} (B) $v_s = 72$ km s^{-1} (C) $v_s = 42$ km s^{-1} (D) $v_s = 62$ km s^{-1}

GR0068

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6. A planet of mass M, has two natural satellites with masses m_1 and m_2. The radii of their circular orbits are R_1 and R_2 respectively. Ignore the gravitational force between the satellites. Define v_1, L_1, K_1 and T_1 to be, respectively, the orbital speed, angular momentum, kinetic energy and time period of revolution of satellite 1 ; and v_2, L_2, K_2 and T_2 to be the corresponding quantities of satellite 2. Given $m_1/m_2 = 2$ and $R_1/R_2 = 1/4$, match the ratios in List-I to the numbers in List-II.

[JEE-Advance 2018]

List–I		**List–II**	
P.	$\frac{v_1}{v_2}$	1.	$\frac{1}{8}$
Q.	$\frac{L_1}{L_2}$	2.	1
R.	$\frac{K_1}{K_2}$	3.	2
S.	$\frac{T_1}{T_2}$	4.	8

(A) P → 4 ; Q → 2 ; R → 1 ; S → 3 (B) P → 3 ; Q → 2 ; R → 4 ; S → 1
(C) P → 2 ; Q → 3 ; R → 1 ; S → 4 (D) P → 2 ; Q → 3 ; R → 4 ; S → 1

GR0069

7. Consider a spherical gaseous cloud of mass density $\rho(r)$ in free space where r is the radial distance from its center. The gaseous cloud is made of particles of equal mass m moving in circular orbits about the common center with the same kinetic energy K. The force acting on the particles is their mutual gravitational force. If $\rho(r)$ is constant in time, the particle number density $n(r) = \rho(r)/m$ is :
[G is universal gravitational constant] **[JEE-Advance 2019]**

(A) $\frac{K}{\pi r^2 m^2 G}$ (B) $\frac{K}{6\pi r^2 m^2 G}$ (C) $\frac{3K}{\pi r^2 m^2 G}$ (D) $\frac{K}{2\pi r^2 m^2 G}$

GR0070

8. The distance between two stars of masses $3M_S$ and $6M_S$ is 9R. Here R is the mean distance between the centers of the Earth and the Sun, and M_S is the mass of the Sun. The two stars orbit around their common center of mass in circular orbits with period nT, where T is the period of Earth's revolution around the Sun. The value of n is ___. **[JEE-Advance 2021]**

GR0108

9. Two spherical stars A and B have densities ρ_A and ρ_B, respectively. A and B have the same radius, and their masses M_A and M_B are related by $M_B = 2M_A$. Due to an interaction process, star A loses some of its mass, so that its radius is halved, while its spherical shape is retained, and its density remains ρ_A. The entire mass lost by A is deposited as a thick spherical shell on B with the density of the shell being ρ_A. If v_A and v_B are the escape velocities from A and B after the interaction process, the ratio $\frac{v_B}{v_A} = \sqrt{\frac{10n}{15^{1/3}}}$. The value of n is _____ **[JEE-Advance 2022]**

GR0198

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	C	D	B	D	C	D	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	B	D	C	D	B	D	C	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	D	D	D	B	A	D	A	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	C	A	D	A	D	C	D	D	D
Que.	41									
Ans.	A									

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	B	A	A	B	A	B	B	D
Que.	11	12	13	14	15	16	17	18	19	
Ans.	B	C	A	B,D	A,B,C	A,D	B,D	B	C	
Que.	20	A	B	C	D					
Ans.		R	Q	Q	P					

EXERCISE # S

1. $\vec{g} = +\frac{\pi G\rho_0 R^3}{6}\left[\frac{1}{(x-(R/2))^2}-\frac{8}{x^2}\right]\hat{i}$, $\vec{g} = -\frac{2\pi G\rho_0 R}{3}\hat{i}$

2. $\frac{R_e k^2}{1-k^2}$ **3.** $h = \frac{\sqrt{5}-1}{2}R$

4. 1.6 hours if it is rotating from west to east, 24/17 hours if it is rotating from east to west

5. 1×10^5J **6.** $t = \frac{GMm}{2C}\left(\frac{1}{R_e}-\frac{1}{r}\right)$ **7.** (a) $T=\frac{2\pi d^{3/2}}{\sqrt{3Gm}}$, (b) 2, (c) 2

8. $L = m\sqrt{[2GMr_1r_2/(r_1+r_2)]}$ **9.** 6

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	4	1	2	4	2	2	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	2	1	3	2	1	2	1	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	4	4	3	4	1	4	4	4
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	2	2	B	D	A	C	A	A	A	B
Que.	41									
Ans.	C									

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9
Ans.	B	2	7	B,C	C	B	D	9	2.30

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